

Alex Chen

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EDUCATION

University of California, Berkeley

Aug 2018 – May 2022

B.S. Computer Science — GPA: 3.82 / 4.0

Relevant Coursework: Database Systems, Machine Learning, Statistical Learning, Data Structures & Algorithms

EXPERIENCE

Stripe

Jul 2022 – Present

Software Engineer II — Payments Infrastructure · San Francisco, CA

- Built real-time fraud signal aggregation pipeline using Kafka + Flink, feeding 40+ ML model features with <200ms end-to-end latency.
- Designed and shipped a high-throughput idempotency layer handling 120K+ transactions/sec using Redis and Postgres, reducing duplicate charges by 99.97%.
- Mentored 2 junior engineers; conducted 50+ code reviews; authored the team's gRPC best-practices guide.

Lyft

May 2021 – Aug 2021

Data Engineering Intern · Remote

- Developed an automated ELT pipeline in Apache Airflow + dbt to ingest and model 8 TB/day of ride-event data into Snowflake.
- Built anomaly detection module using Isolation Forest, flagging 94% of GPS spoofing incidents before driver review.
- Reduced Spark job runtime by 62% via partition pruning and broadcast join optimizations across 15 production DAGs.
- Created interactive Tableau dashboards for driver supply/demand forecasting, adopted by 3 city operations teams.

Accenture

Jun 2020 – Aug 2020

Technology Analyst (Frontend) · Remote

- Implemented responsive layouts and WCAG 2.1 AA accessibility improvements, reducing reported a11y issues by 78%.

PROJECTS

NeuralResume

github.com/alexchen-dev/neural-resume

Tech: Python, PyTorch, HuggingFace, FastAPI, PostgreSQL

Fine-tuned a BERT-based NLP classifier on 12K labeled job descriptions to extract skills and match candidates. Achieved F1=0.91 on held-out test set. Packaged as a FastAPI microservice.

PulseBoard

github.com/alexchen-dev/pulseboard

Tech: React, TypeScript, D3.js, WebSockets, FastAPI

Real-time analytics dashboard with D3.js visualizations, WebSocket-fed live metrics, and a drag-and-drop widget layout. Used by 300+ users in open beta.

StreamQL

github.com/alexchen-dev/streamql

Tech: Java, Kafka Streams, Avro, Maven

Lightweight streaming SQL engine built on top of Kafka Streams that supports windowed aggregations and joins with sub-second latency.

SKILLS

Languages: Python, Java, SQL, R

Frameworks: PyTorch / TensorFlow, Apache Spark / Flink, Kafka Streams, D3.js / Three.js

Tools: Apache Airflow / dbt, GitHub Actions / Jenkins, Git

Databases: PostgreSQL / MySQL, Snowflake / BigQuery, Elasticsearch

ACHIEVEMENTS & CERTIFICATIONS

- Google Professional Data Engineer

Google Cloud 2022

- Published: 'Efficient Approximate Nearest Neighbor Search in High Dimensions' — ACM KDD 2022

ACM 2022

- Speaker: PyCon US 2023 — 'Building Fraud Signals at Scale with Kafka and Python'

Python Software Foundation 2023

- HackMIT 2021 — Best Infrastructure Hack

MIT 2021